

International Islamic University Chittagong
Faculty of Shariah and Islamic Studies
Department of Qur'anic Sciences and Islamic Studies

Semester End Examination : Autumn 2025

Program: B A (Hon's)

Course Code: STAT-3503

**Course Title: Statistics for Research in
Shariah Studies**

Time: 2 hours and 30 minutes

Full Marks: 50

[Answer the following questions. Figures in the right margin indicate full marks.]

QN	Description of Questions	Marks	CLOs & PLOs	Cognitive learning														
1(a).	Identify the different measures of Central tendency? Arithmetic Mean is affected by extreme values- explain this statement? ✓	3	CLO1, PLO3	Remembering Analyzing														
1(b).	The following data are the figures of profit earned by 1400 companies during 2024-2025 fiscal years. <table><tr><th>Profits (Taka Lakhs)</th><th>No. of Companies</th></tr><tr><td>200-400</td><td>500</td></tr><tr><td>400-600</td><td>300</td></tr><tr><td>600-800</td><td>280</td></tr><tr><td>800-1000</td><td>120</td></tr><tr><td>1000-1200</td><td>100</td></tr><tr><td>1200-1400</td><td>80</td></tr><tr><td>1400-1600</td><td>20</td></tr></table> Calculate the average profits for all the companies by using short-cut method.	Profits (Taka Lakhs)			No. of Companies	200-400	500	400-600	300	600-800	280	800-1000	120	1000-1200	100	1200-1400	80	1400-1600
Profits (Taka Lakhs)	No. of Companies																	
200-400	500																	
400-600	300																	
600-800	280																	
800-1000	120																	
1000-1200	100																	
1200-1400	80																	
1400-1600	20																	
2(a).	Write the properties of an ideal measure of Variation ✓	3	CLO2, PLO3 PLO4	Analyzing, Understanding														
2(b).	Imagine you and your classmates are the owners of a small bakery. Some days the sales are high, and other days they're low. As the owners, you want to understand the sales trends so you can better plan for the future and make smarter decisions about stocking up on ingredients. <table><tr><th>Sales (in thousand Taka)</th><th>No. of days</th></tr><tr><td>10-20</td><td>3</td></tr><tr><td>20-30</td><td>6</td></tr><tr><td>30-40</td><td>11</td></tr><tr><td>40-50</td><td>3</td></tr><tr><td>50-60</td><td>2</td></tr></table> As the bakery owners, calculate the average deviation from the mean of the sales.	Sales (in thousand Taka)			No. of days	10-20	3	20-30	6	30-40	11	40-50	3	50-60	2	7		
Sales (in thousand Taka)	No. of days																	
10-20	3																	
20-30	6																	
30-40	11																	
40-50	3																	
50-60	2																	
OR																		
2(c)	What do you mean by range, average deviation and standard deviation? Explain their importance properties	10	CLO2, PLO 3	Remembering														

3(a).	You are conducting a study to evaluate the impact of online learning on students' performance at IIUC, with a total of 5,000 students from five faculties. You need to select 500 students for your survey. i) If you use Stratified Sampling, how would you decide how many students to pick from each faculty? Why is this method useful for your study? ✓ ii) ✓ Which of the four sampling methods would you choose for this study? Explain why, considering time, cost, fairness, and how well the sample represents all students.	5+5	CLO4, PLO2	Analyzing , Understanding																				
OR																								
3(b).	What is sampling? Describe the four probability sampling methods.	5	CLO2, PLO2	Understanding																				
3(c).	Write the essentials of sampling and principal step in a sample survey.	5																						
4 ✓	The following data give the test scores and sales made by nine salesmen during the last one year: <table><tr><td>Test scores</td><td>14</td><td>19</td><td>24</td><td>21</td><td>26</td><td>22</td><td>15</td><td>20</td><td>19</td></tr><tr><td>Sales</td><td>31</td><td>36</td><td>48</td><td>37</td><td>50</td><td>45</td><td>33</td><td>41</td><td>39</td></tr></table> Obtain I) the regression equation of test scores on sales; II) find test scores if a salesman's sale is 55; <i>-67.8478</i> III) the regression equation of sales on test scores; IV) find sales if test scores is 30; <i>378.2016</i> V) coefficient of correlation. <i>0.557, 1.608</i>	Test scores	14	19	24	21	26	22	15	20	19	Sales	31	36	48	37	50	45	33	41	39	10	CLO3, PLO2	Analyzing
Test scores	14	19	24	21	26	22	15	20	19															
Sales	31	36	48	37	50	45	33	41	39															
5 ✓	Explain Linear Regression Models, panel data models, Time Series Models. Discuss the advantages of Econometrics?	10	CLO3, PLO4	Remembering																				

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Full marks: 50

Bloom's Levels of the Questions						
Bloom's Meaning	Create	Evaluate	Analyze	Apply	Understand	Remember

[Note: Answer all questions. Figures in the right margin indicate full marks. Answer should be brief, to the point and relevant. Parts of each question should be answered consecutively]

Serial No	Questions		Marks	Bloom's level	CLO																		
1	a	Define range in a set of data. What purpose does a measure of variation serve in research?	2	Understand	CLO-1																		
	b	A survey of domestic consumption of electricity gave the following current consumed: <table><tr><th>Number of units</th><th>No. of consumers</th></tr><tr><td>0-200</td><td>9</td></tr><tr><td>200-400</td><td>18</td></tr><tr><td>400-600</td><td>27</td></tr><tr><td>600-800</td><td>32</td></tr><tr><td>800-1000</td><td>45</td></tr><tr><td>1000-1200</td><td>38</td></tr><tr><td>1200-1400</td><td>20</td></tr><tr><td>1400-1600</td><td>2</td></tr></table> Compute average deviation of consumption from median and mode.	Number of units	No. of consumers	0-200	9	200-400	18	400-600	27	600-800	32	800-1000	45	1000-1200	38	1200-1400	20	1400-1600	2	8	Apply	CLO-2
Number of units	No. of consumers																						
0-200	9																						
200-400	18																						
400-600	27																						
600-800	32																						
800-1000	45																						
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1200-1400	20																						
1400-1600	2																						
Or of 1	a	What do you mean by range, average deviation and standard deviation? Explain their important properties.	10	Understand	CLO-3																		
2	a	What is sampling? What are the justifications for selecting a sample in research?	5	Remember	CLO-2																		
	b	Enumerate the various methods of sampling and principal step in a sample survey.	5	Understand	CLO-2																		

3	a	Distinguish between parameter and statistic. Suppose you are asked to conduct a survey on the socio-economic background of the IIUC students. How will you proceed?	6	Evaluate	CLO-4																
	b	Suppose a sample of 100 students is to be drawn from a population consisting of 1000 students belonging to three departments QSI, SHIS & DIS under the Faculty of Shariah and Islamic Studies at IIUC. The number of students in the department of QSI is 400, SHIS is 300 and the DIS is 300. How will you draw the sample using proportional allocation?	4	Apply	CLO-4																
4		What do you mean by case study? Discuss the case study research steps and methods with examples.	10	Understand	CLO-2																
Or of 4		What does regression analysis examine? Consider the price(P) and Supply(S) information as below:	10	Analyze	CLO-4																
<table border="1"> <tr> <td>P</td> <td>1</td> <td>2</td> <td>3</td> <td>4</td> <td>5</td> <td>6</td> </tr> <tr> <td>S</td> <td>4</td> <td>6</td> <td>7</td> <td>7</td> <td>9</td> <td>11</td> </tr> </table> Estimate the supply function and find out the intercept and slope.						P	1	2	3	4	5	6	S	4	6	7	7	9	11		
P	1	2	3	4	5	6															
S	4	6	7	7	9	11															
5	a	What do you mean by median and mode?	2	Remember	CLO-1																
	b	The following data relate to the sales of 100 companies,	8	Apply	CLO-3																
<table border="1"> <tr> <th>Sales (Tk. Lakhs)</th> <th>No. of companies</th> </tr> <tr> <td>Below 60</td> <td>12</td> </tr> <tr> <td>60-62</td> <td>18</td> </tr> <tr> <td>62-64</td> <td>25</td> </tr> <tr> <td>64-66</td> <td>30</td> </tr> <tr> <td>66-68</td> <td>10</td> </tr> <tr> <td>68-70</td> <td>3</td> </tr> <tr> <td>70-72</td> <td>2</td> </tr> </table> Calculate the median and modal sales of companies.						Sales (Tk. Lakhs)	No. of companies	Below 60	12	60-62	18	62-64	25	64-66	30	66-68	10	68-70	3	70-72	2
Sales (Tk. Lakhs)	No. of companies																				
Below 60	12																				
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THE END